

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks: 20

Annexure A

Descriptive Written Test

Code of Conduct:

*I Dharshan B (192312150) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Dharshan

Level-3
[Signature]

Introduction:

A smart home automation system controls lighting, temperature and security using sensors such as motion detectors, temperature sensors and clocks.

Types of Intelligent Agent:

1. Simple Reflex Agent

- * Acts only on the current sensor input using preprogrammed rules.
- * Advantages: Fast, simple, low cost.
- * Disadvantages: cannot remember past events or learn user preferences.

2. Model-Based Agent

- * Maintains an internal model (memory) of the environment.
- * Example:
 - * knows whether someone is already in a room even if motion temporarily stops.
 - * Maintains room status (occupied/unoccupied).
- * Advantages: Better decisions in partially observable environments.
- * Disadvantages: More memory and computation required.

3. Goal-Based Agent

- * Makes decisions to achieve specific goals.
- * Goals:
 - maintain comfortable room temperature.
 - secure the house.
 - Reduce electricity usage.
- * Advantages: Flexible and can choose the best sequence of action.

* Disadvantages: planning increases computation time

4. Utility-Based Agent

* selects the action that gives the highest overall benefit.

* considers multiple factors simultaneously:

- user comfort
- Energy consumption
- security

* Example:

- Adjusts AC to 25°C instead of 20°C to balance comfort and electricity savings.

* Advantages: produces the best overall decision

* Disadvantages: utility calculation is complex

Best Approach:- Hybrid Agent

* Model-Based remembers room status and user activities

* Utility-Based balances comfort, energy efficiency and security

* Learning capability adapts to user preferences over time

Conclusion:

A hybrid intelligent agent provides the best smart home automation by ensuring user comfort, reducing energy consumption, improving security, and continuously learning user habits.

3) Smart Traffic Management System:

Introduction:-

A Smart traffic management system controls traffic lights using vehicle density, pedestrian movement, and weather conditions to reduce congestion and travel time.

Types of intelligent Agents

1. Simple Reflex Agent

- * Uses immediate sensor data

- * Example:

 - Heavy traffic $\rightarrow$ Green Signal

 - No vehicle $\rightarrow$ Red Signal

- * Advantage: very fast

- * Disadvantage: cannot learn traffic patterns

2. Model - Based Agent

- * Maintains information about traffic flow

- * Example:

 - Remembers congestion at nearby intersections

- * Advantage: Better coordination

- * Disadvantage: Requires more processing

3. Goal - Based Agent

- * A Goal - Based Agent chooses actions that help achieve pre-defined goals.

- Goals:-

- * Reduce vehicle waiting time

- * minimize traffic congestion
- * Ensure pedestrian safety
- * Give priority to emergency vehicles
- * Advantages: Flexible decision-making, can plan future actions
- * Disadvantage: Planning increases computational complexity

4. Utility-Based Agent:

Factors Considered

- * Vehicle waiting time
- * Fuel consumption
- * Emergency vehicle priority
- * pedestrian safety
- * Traffic congestion

Example :- During heavy rain:

- * Increase pedestrian crossing time
- * Reduce vehicle speed through longer yellow signal
- * Give priority to emergency vehicles

* Advantage:-

- * Produce optimal decisions

Disadvantage:- utility calculation is complex.

Conclusion:-

A hybrid intelligent agent provides efficient traffic patterns and adjusts by reducing congestion, minimizing travel time, improving fuel efficiency, and adapting traffic conditions through continuous learning.